Band-1 Kakeya / Restriction: Next Dimension or Endpoint (Collapse 2.09)

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Band-1 Kakeya / Restriction — Lit + Statement Lock

Honesty stamp. This note does **not** claim the Kakeya set conjecture in **all** dimensions. It does **not** claim the Stein restriction conjecture (endpoint) solved in $\mathbb{R}^3$. It does **not** claim a Clay-millennial solution or a Theory of Everything. **GATE-04** is locked out of scope.

> **Locked:** 3D Kakeya **set** conjecture — **PROVED** (Wang–Zahl 2025).
> **Locked:** Kakeya in $n \geq 4$; 3D Kakeya **maximal function**; 3D **restriction endpoint** — **OPEN**.

Abstract

Literature heat (2017–2026): Hong Wang and Joshua Zahl proved the **Kakeya set conjecture in** $\mathbb{R}^3$: every Kakeya set in three dimensions has Hausdorff and Minkowski dimension 3 (arXiv:2502.17655, Feb 2025), completing a trilogy that began with the sticky case (arXiv:2210.09581). A streamlined sticky→full reduction appears as Guth–Wang–Zahl (arXiv:2601.14411, 2026). Tao's contemporaneous exposition and the IAS Letter (Spring 2025) emphasize that the stronger **Kakeya maximal-function** conjecture in $\mathbb{R}^3$ remains open, and that dimensions $n \geq 4$ remain open. The **restriction** tower in $\mathbb{R}^3$ (Stein) is likewise **not** closed; Guth's polynomial-partitioning extension bound $p > 3.25$ remains the landmark partial result short of the endpoint.

Box: optional 2D tube-union area cartoon only (Davies already closed $n = 2$); **not** a 3D or restriction proof.

Problem (collapse Band-2 item 2.09 → Band-1 lit+statement)

1. Ledger what is **proved** vs **open** for Kakeya / restriction after the 2025 Wang–Zahl breakthrough.
2. Name the **next** targets: next dimension ($n = 4$) and/or quantitative endpoints (maximal Kakeya; restriction $p = 3$ in $\mathbb{R}^3$).
3. Keep all-dimension Kakeya, Clay, ToE, and GATE-04 off the claim surface.

Literature ledger (web check 2017–2026; not re-proved on box)

Year	Result	Authors	Status
1971	Kakeya set conj. in $\mathbb{R}^2$	Davies	classical **closed**
2016	Restriction/extension $p > 3.25$ in $\mathbb{R}^3$ (poly. partitioning)	Guth, *JAMS* 29	**partial**; endpoint **OPEN**
2019	Hausdorff dim Besicovitch in $\mathbb{R}^3$ $\geq 5/2 + \varepsilon$	Katz–Zahl, *JEMS*	historical; **superseded** for set conj. by WZ 2025

Year	Result	Authors	Status
2022	**Sticky** Kakeya theorem in $\mathbb{R}^3$	Wang–Zahl, arXiv:2210.09581	**proved** (special case)
~2022–24	Assouad-dim / structural middle paper (trilogy)	Wang–Zahl	intermediate **proved**
2025	**Kakeya SET conjecture in** $\mathbb{R}^3$ (Hausdorff + Minkowski dim = 3)	**Wang–Zahl**, arXiv:2502.17655	**PROVED** (breakthrough preprint)
2025	Exposition: set closed; maximal still open	Tao, *What's new* (2025-02-25)	secondary
2025	Survey of Kakeya 2000–2025	Zahl et al. / SIAM survey tracks	secondary
2026	Streamlined sticky→full proof in $\mathbb{R}^3$	Guth–Wang–Zahl, arXiv:2601.14411	streamlined
—	Kakeya set conj. in $n \geq 4$	—	**OPEN**
—	Kakeya **maximal function** conj. in $\mathbb{R}^3$	—	**OPEN**
—	Stein **restriction** endpoint in $\mathbb{R}^3$ ($n \geq 3$)	—	**OPEN**

Canonical 3D set citation. H. Wang, J. Zahl, *Volume estimates for unions of convex sets, and the Kakeya set conjecture in three dimensions*, arXiv:2502.17655 (2025).

Canonical sticky citation. H. Wang, J. Zahl, *Sticky Kakeya sets and the sticky Kakeya conjecture*, arXiv:2210.09581.

Restriction landmark (partial). L. Guth, *A restriction estimate using polynomial partitioning*, Journal of the American Mathematical Society **29** (2016), 371–413.

Hygiene. "Hong–Zahl" in dash shorthand = **Hong Wang + Joshua Zahl** (Wang–Zahl). Do **not** elevate media phrasing ("Kakeya settled") to all dimensions or to the maximal/restriction endpoints. Preprint peer-review status of arXiv:2502.17655 may evolve; the mathematical claim of the 3D **set** conjecture is treated as the standing breakthrough record for this ledger.

What is proved vs open (statement lock)

Proved (cite carefully)

1. $\mathbb{R}^2$ **Kakeya set conjecture** — Davies (classical).
2. **Sticky Kakeya in** $\mathbb{R}^3$ — Wang–Zahl (arXiv:2210.09581): sticky configurations have full dimension 3.
3. **Full Kakeya SET conjecture in** $\mathbb{R}^3$ — Wang–Zahl (arXiv:2502.17655): every Kakeya set in $\mathbb{R}^3$ has Minkowski and Hausdorff dimension 3. Consequence of near-maximal volume estimates for δ-tubes under convex Wolff / non-clustering axioms.
4. **Partial restriction** — Guth (2016): extension estimates for positively curved surfaces in $\mathbb{R}^3$ for $p > 3.25$, short of the conjectured endpoint.

Open (next dimension / endpoint)

1. **Kakeya set conjecture in** $n \geq 4$ — **OPEN**. Wang–Zahl methods lean on 3D geometry; higher dimensions retain only partial lower bounds.
2. **Kakeya maximal-function conjecture in** $\mathbb{R}^3$ — **OPEN**. Quantitative tube-overlap control stronger than the set-dimension statement; Wang–Zahl yield nontrivial bounds but do not close it (Tao 2025; IAS 2025).

3. **Stein restriction conjecture endpoint in** $\mathbb{R}^3$ (and $n \geq 3$) — **OPEN**. Closely linked to maximal Kakeya / Fourier restriction; Guth $p > 3.25$ is not the endpoint.
4. Related **local smoothing** tower pieces in $\mathbb{R}^3$ — still largely open at the sharp endpoint.

Next named targets (collapse 2.09)

Priority	Target	Type
A	Kakeya set conjecture in $\mathbb{R}^4$	**next dimension**
B	Kakeya maximal function conjecture in $\mathbb{R}^3$	**quantitative endpoint**
C	Restriction endpoint $p = 3$ in $\mathbb{R}^3$	**Fourier endpoint**

Optional box illustration (not a proof)

2D discrete tube-union cartoon on a 256×256 grid: $N = 48$ directions, width $\delta = 0.04$, random segment centers in $[0.2, 0.8]^2$.

Quantity	Value
Union area fraction	≈ 0.732
Naive sum-of-areas (capped)	1.0
Label	overlap cartoon only; **Davies already closed** $n = 2$; **not** 3D Kakeya; **not** restriction

Why Band-1 vs conjecture

- **Band-1 (this probe):** honest lit+statement lock after Wang–Zahl 2025; name next dimension/endpoint; optional tiny 2D cartoon.
- **Not claimed:** Kakeya in all dimensions; 3D maximal Kakeya; restriction endpoint; Clay; ToE.
- **GATE-04:** LOCKED.

Reproducibility

- Lab: `/workspace/mission-control/experiments/band1/kakeya_restriction/`
- Artifacts: `literature_ledger.json`, `results.json`, `SUMMARY.txt`, `METHOD.md`, `README.md`
- Optional cartoon: `python3 -c` recreatable from `results.json` parameters (numpy grid mask)

Conclusion

Lit lock: Wang–Zahl (2025) **prove** the Kakeya **set** conjecture in $\mathbb{R}^3$; sticky precursor Wang–Zahl (2022); Guth–Wang–Zahl (2026) streamline. **Open:** Kakeya $n \geq 4$; 3D Kakeya **maximal function**; 3D **restriction endpoint** (Guth $p > 3.25$ partial).
Box: optional 2D union cartoon only.
Do not claim Kakeya solved in all dimensions. No Clay / ToE. GATE-04 locked.